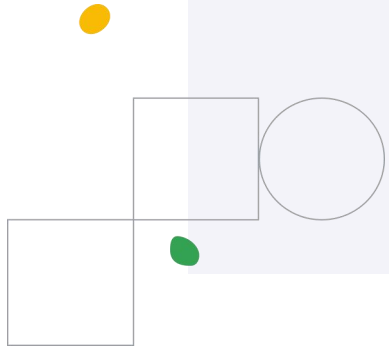
Foundations of Generative AI

01



Learning objectives

Define generative AI

Explain how generation differs from prediction

Compare major model families

Connect applications to business needs

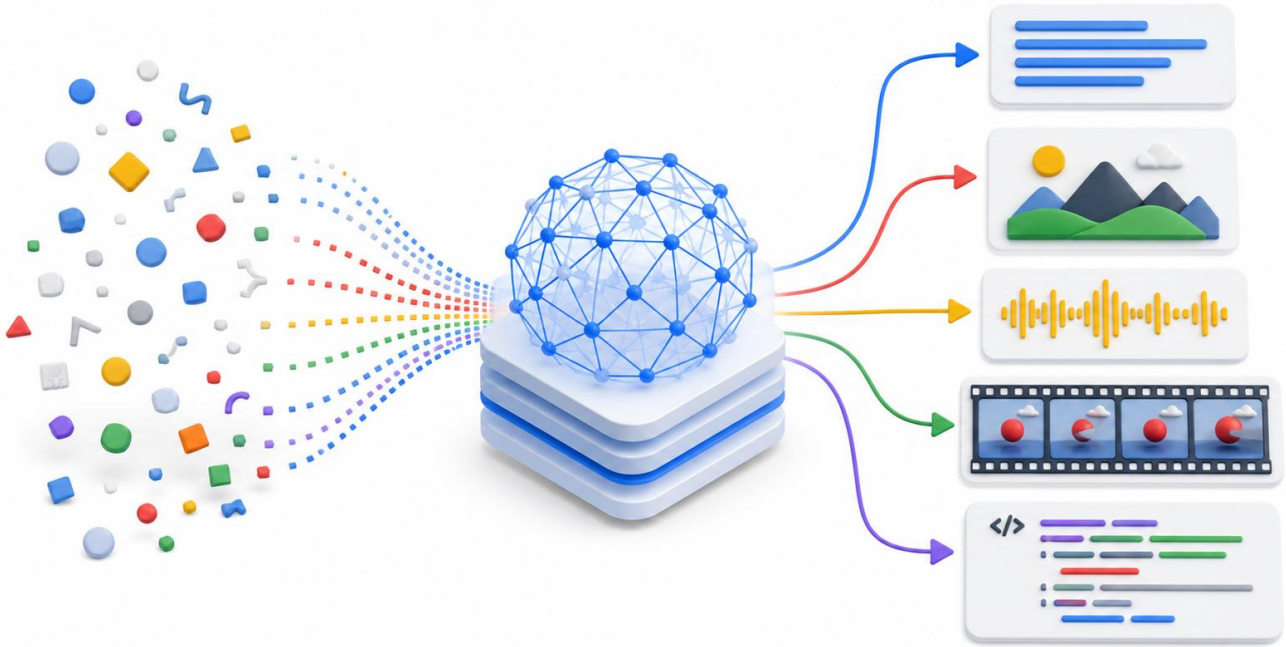
Module topics





Foundations of Generative AI

From prompt to generated content



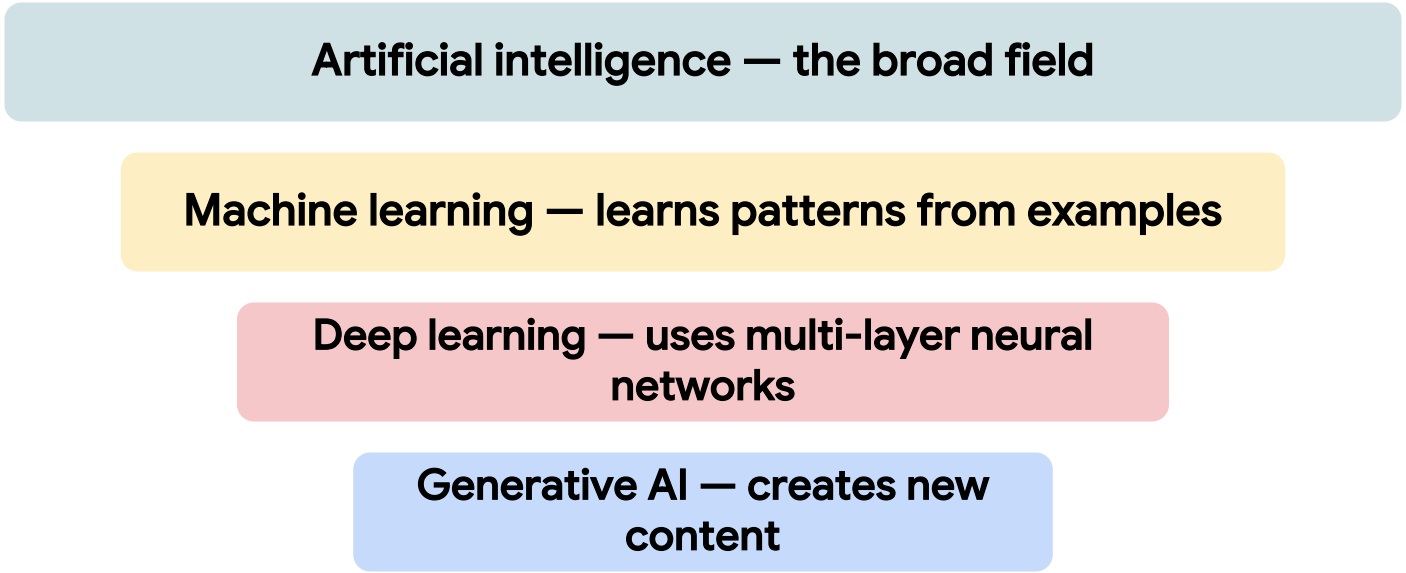
Generative AI creates content from learned patterns

Training examples reveal patterns in language, code, images, audio, and video.

The model uses those patterns to create a new output for the user’s request.

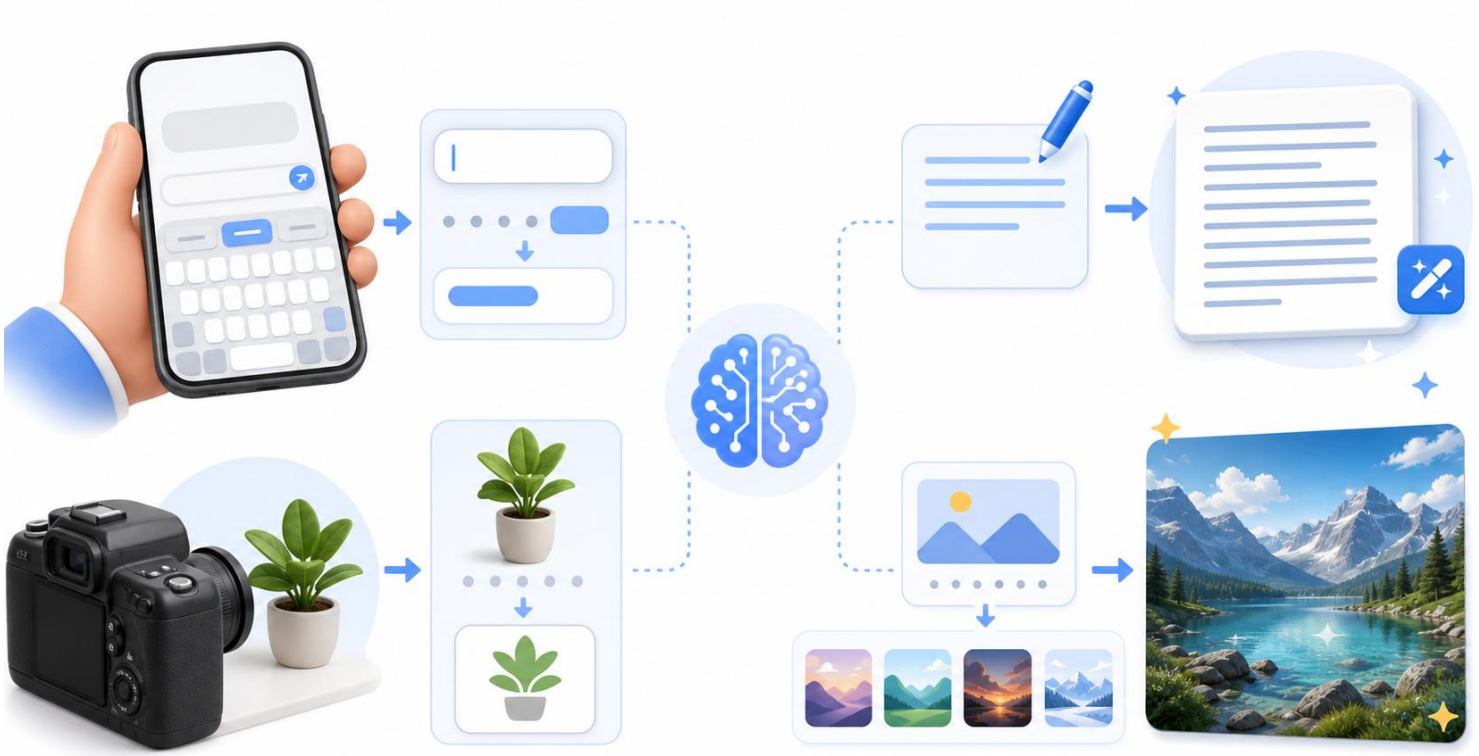
Generated content is new, but it is shaped by what the model learned.

AI, ML, deep learning, and Gen AI form a hierarchy

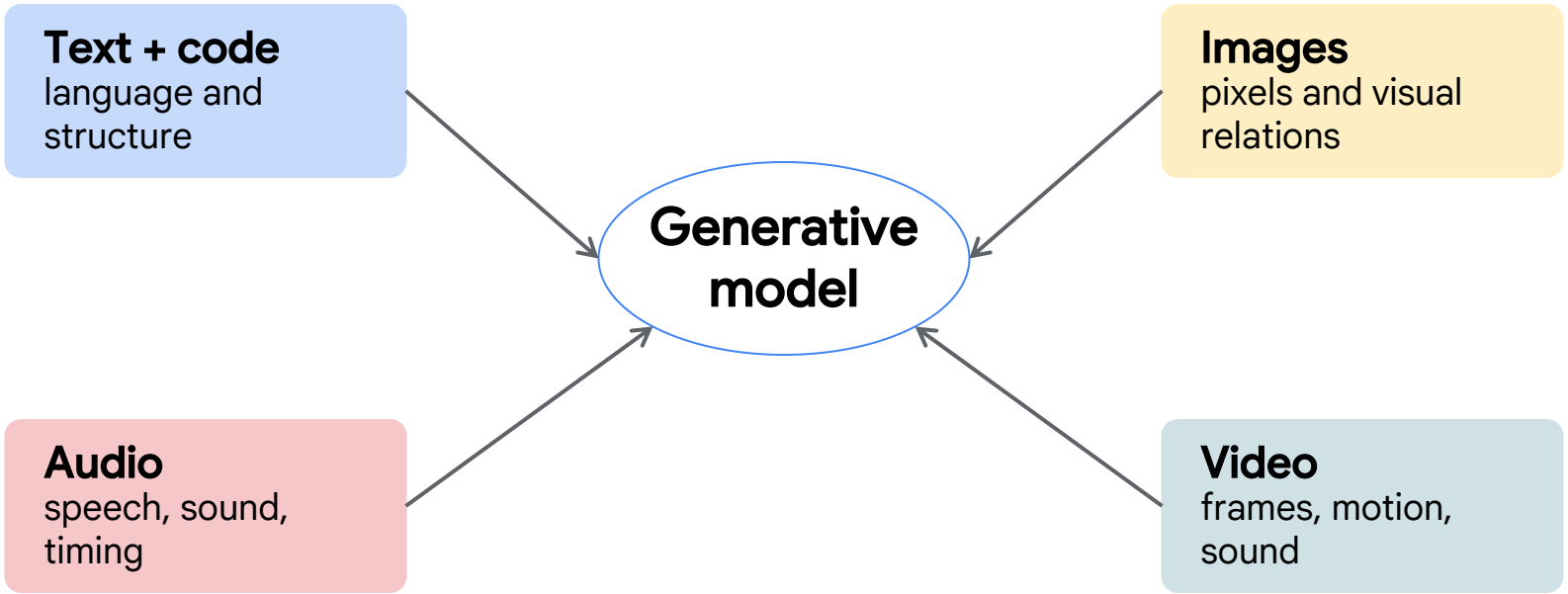


Each lower layer is a more specialized subset of the layer above.

Familiar tools can combine prediction and generation



Generative systems learn from several forms of data



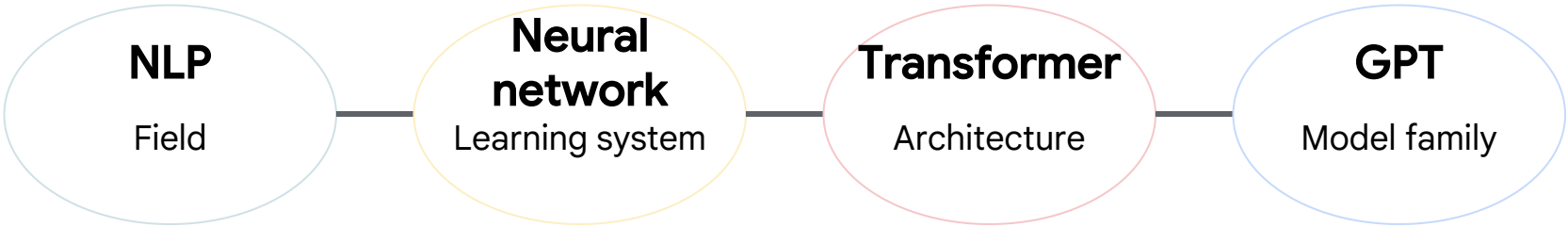
A neural network learns by adjusting connections

Information passes through layers of connected mathematical units.

Training compares the answer with the expected result and measures the error.

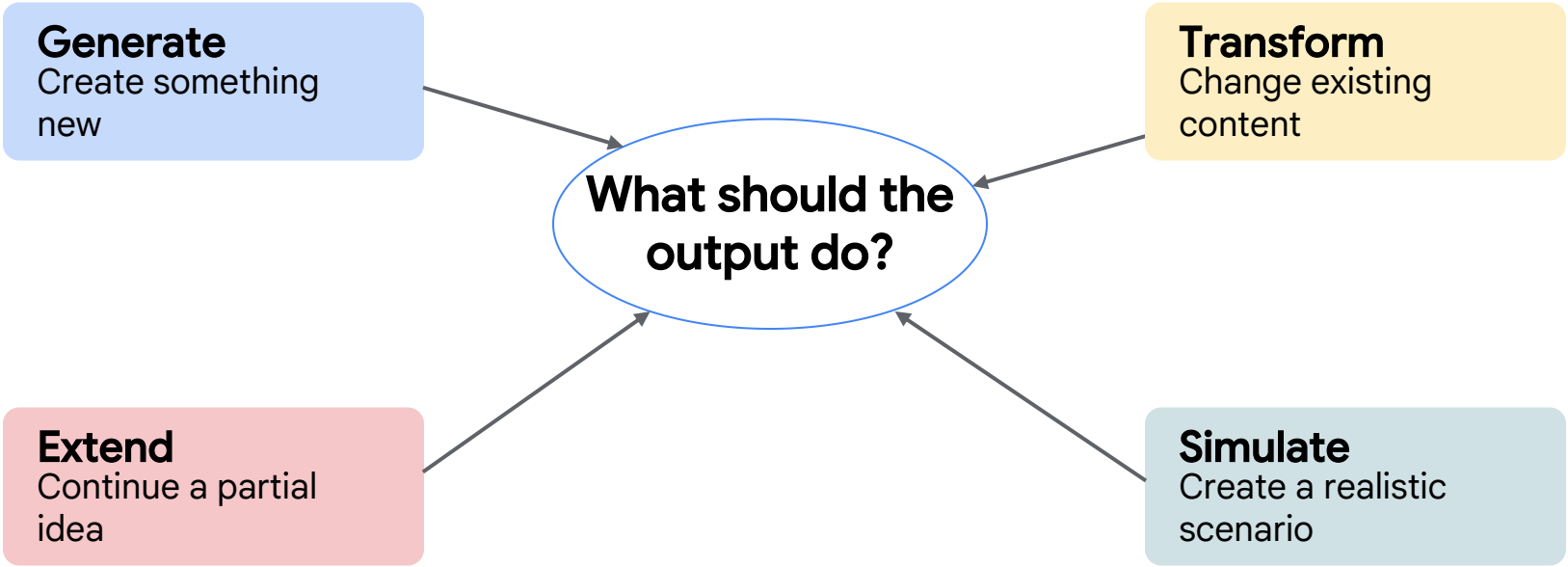
The network adjusts its connection weights so later answers improve.

Language AI connects NLP, neural networks, and GPT



NLP is the problem area; neural networks learn; Transformers organize attention; GPT names a generative Transformer family.

Generative AI supports four basic task patterns



Traditional ML predicts; Generative AI produces

Traditional ML classifies, detects, recommends, or forecasts.

Generative AI creates a draft, response, image, program, or other new content.

Example: classify a support ticket—or generate a reply to that ticket.

One dataset can support prediction and generation

Support conversations can train a predictive model to estimate urgency.

The same type of data can help a generative model summarize or draft a response.

The task definition determines what the system is designed to produce.

Different model families generate different kinds of content

Transformers	Diffusion	GANs	VAEs
Context + sequences	**Iterative denoising**	**Adversarial competition**	**Structured latent space**
Text, code, multimodal	Images and video	Realistic synthesis	Variation + anomalies

GANs learn through competition

A generator creates artificial examples; a discriminator tries to detect the fakes.

As both networks improve, the generator learns to create increasingly realistic output.

Use cases: image synthesis, restoration, super-resolution, and data augmentation.

Example: create varied product images or rare training examples for a vision system.

VAEs learn a structured latent space

An encoder compresses an example into a probability distribution in a latent space.

A decoder samples that space to reconstruct the input or generate a related variation.

Use cases: representation learning, anomaly detection, interpolation, and controlled generation.

Example: move between two product designs to generate meaningful intermediate variations.

Choose the model family that matches the task

Compare	Transformer	Diffusion	GAN	VAE
Core idea	Attention	Remove noise	Compete	Encode + decode
Best known for	Language + context	High-quality visuals	Photorealism	Controlled variation
Example	Help-desk reply	Course illustration	Synthetic defect image	Product interpolation

Transformers are designed to use context

A Transformer examines relationships among different parts of an input.

Attention helps it focus on the words, symbols, or regions that matter most.

This makes Transformers useful for language, code, images, audio, and video.

Diffusion models learn to remove noise

Training teaches the model to recover useful structure from noisy examples.

Generation begins with noise and repeatedly refines it toward the prompt.

This approach is widely used for creating and editing images.

Training and inference are two distinct phases

Training: the model processes large datasets, predicts targets, measures error, and updates parameters.

Inference: a user supplies a prompt; the trained model generates an output without retraining.

Feedback, grounding, tuning, and evaluation can improve application quality.

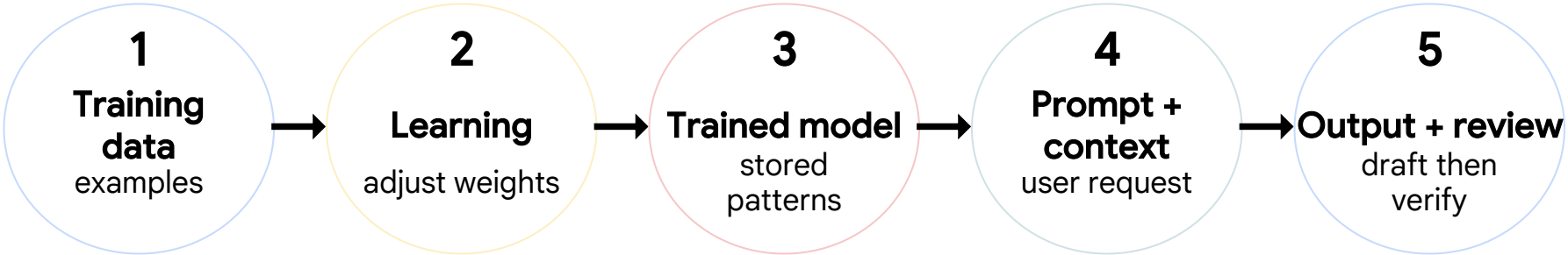
A trained model stores parameters—not a copy of every lesson

Training compresses statistical relationships into numerical weights.

During use, the model calculates an output from those weights and the prompt.

Privacy and testing still matter because models can sometimes reproduce fragments.

Generation follows a repeatable four-step flow



Four inputs shape what a generative model returns

1	**Instruction**	What to do
2	**Context**	What the model should know
3	**Examples**	What good output resembles
4	**Constraints**	Rules, tone and format

A clearer prompt reduces ambiguity and makes the output easier to evaluate.

Generative AI is probabilistic—not a database lookup

The model calculates likely outputs from the prompt and available context.

Different settings can produce several reasonable answers to the same request.

Important claims must still be verified against trusted information.

One request can produce several valid answers

A welcome message may emphasize belonging, academic opportunity, or support.

This variation is useful for creative work and exploring alternatives.

Repeatable tasks need tighter instructions, controlled settings, and validation.

Example: a university help-desk assistant

Input: a student asks how to replace a lost identity card.

Generation: the model drafts a response using approved university policy.

Validation: the application checks fees, deadlines, and policy references.

Human support: unusual or high-impact cases are escalated to staff.

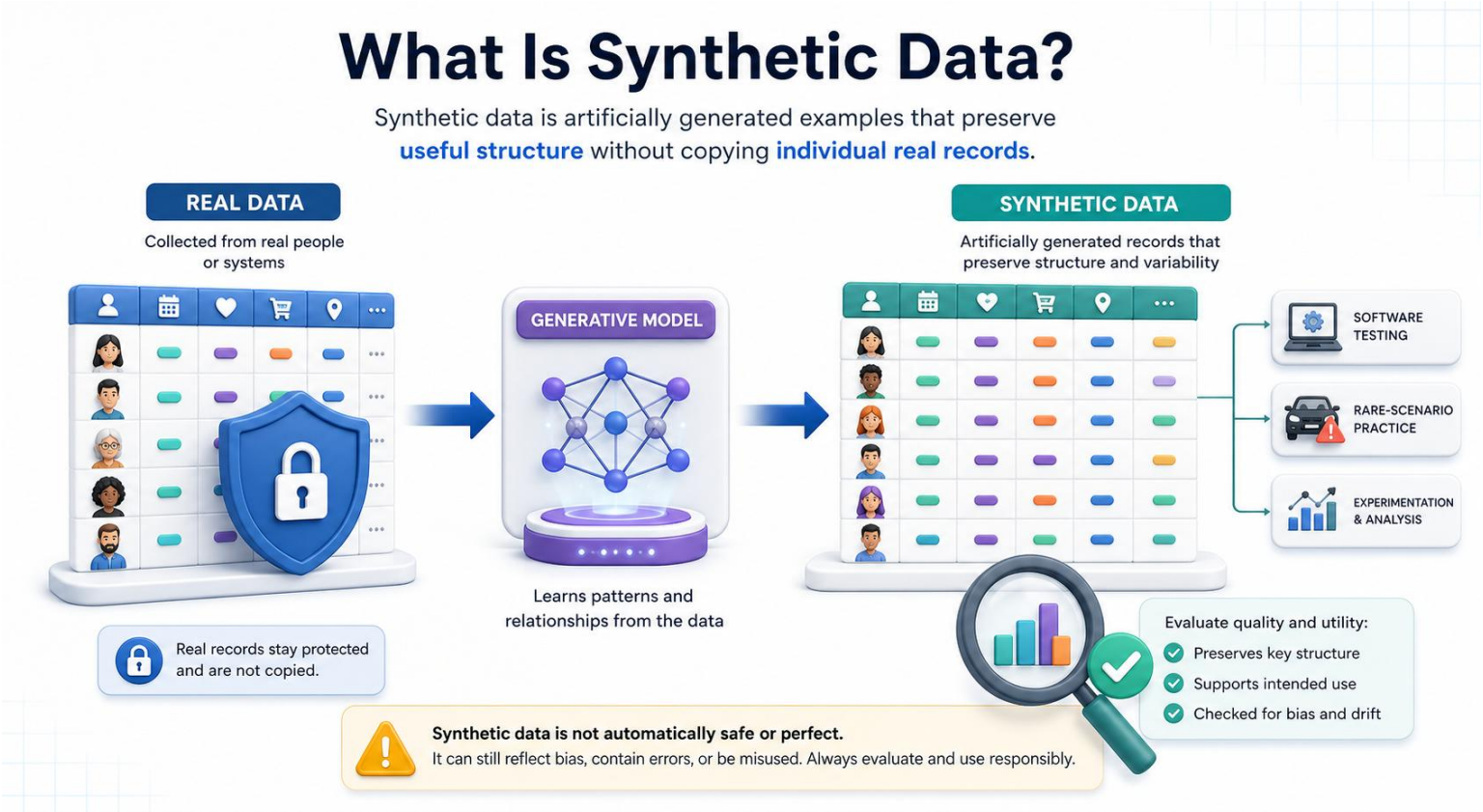
Education use cases support teaching, access, practice, and feedback



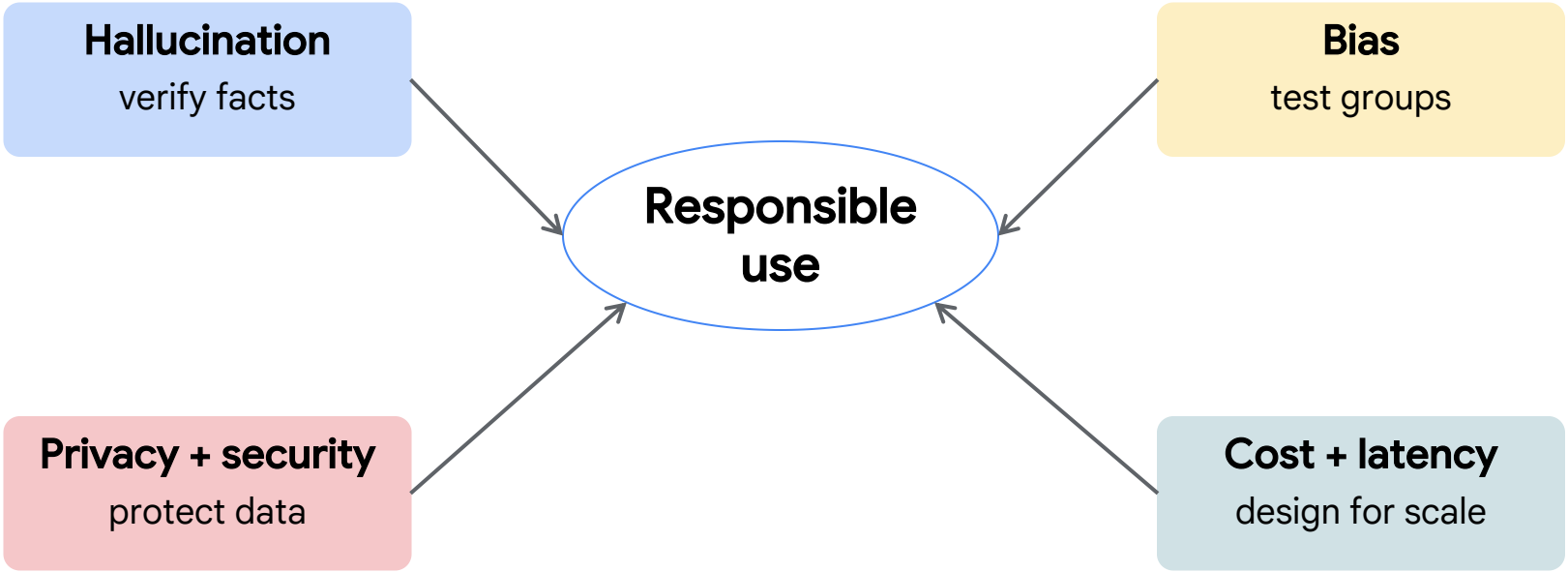
Organizations use Gen AI to assist common work



Synthetic data is useful only when quality is evaluated



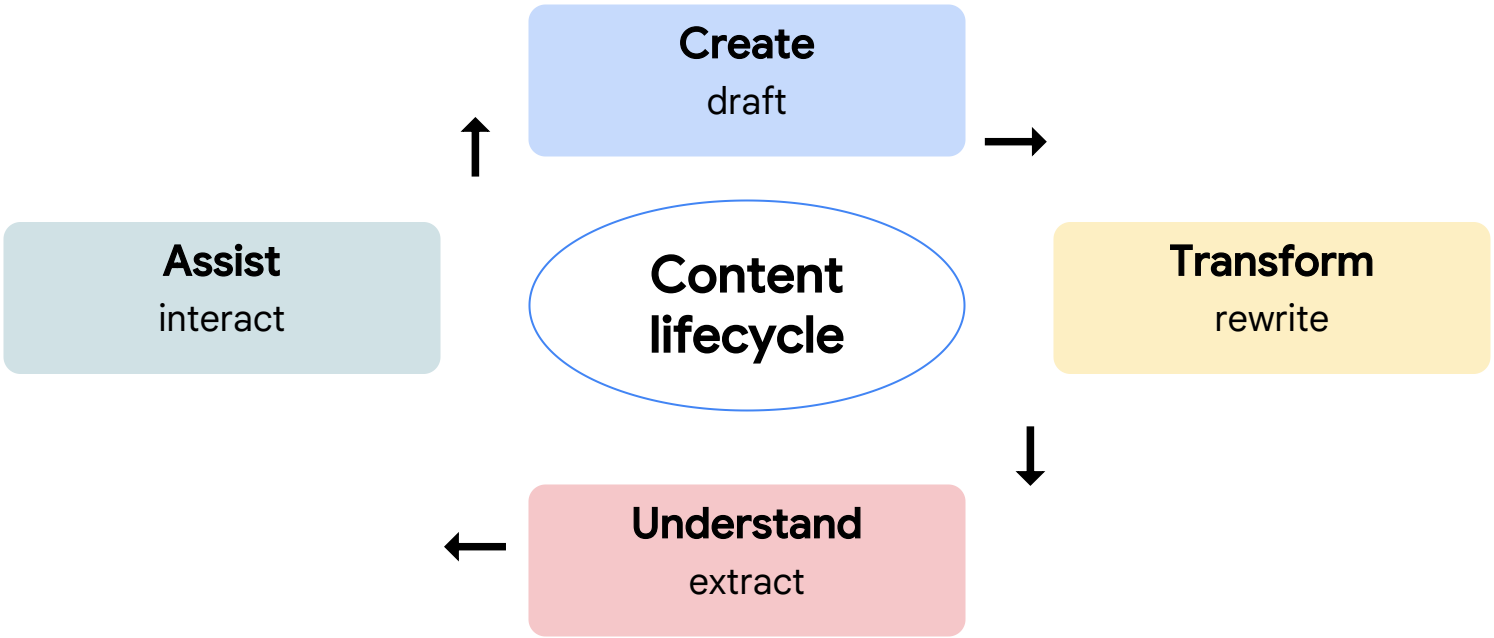
Four limitations shape responsible use



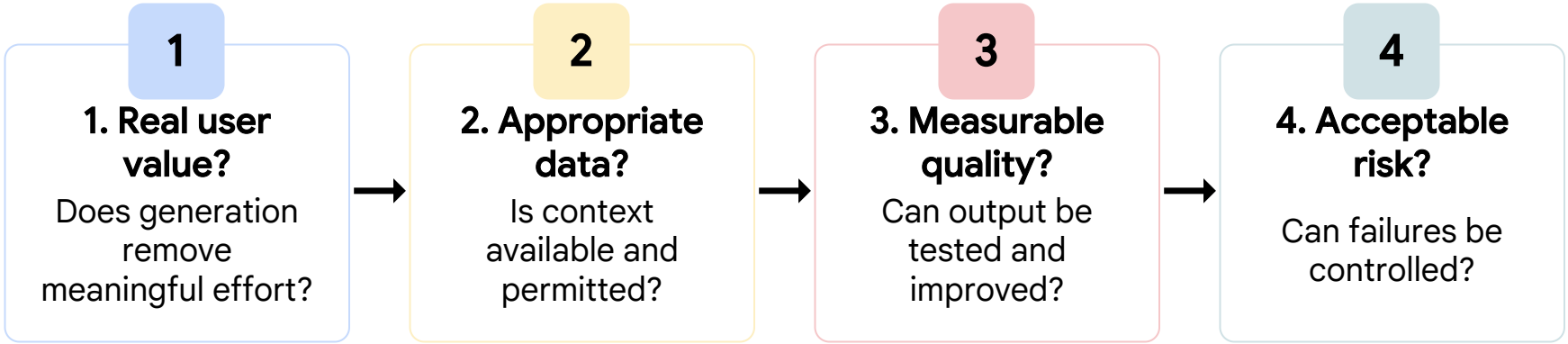
Common misconceptions can lead to poor decisions

MYTH	REALITY
It understands like a person	It predicts patterns; fluent language is not human understanding
A confident answer is correct	Fluency and factual accuracy are different
More data solves every problem	Quality, relevance, rights and representation matter
The model owns the decision	People and organizations remain accountable

Applications span the full content lifecycle



A simple suitability check starts with four questions



Four ‘yes’ answers justify a controlled pilot—not automatic deployment.

Use Gen AI for flexible content and interaction

Good fit: drafting, summarization, ideation, search, and content transformation.

Use safeguards for exact facts, arithmetic, and high-impact decisions.

Begin with the user problem; then select and evaluate the model.

Skills Google

Get Started with Vertex AI Studio (GSP1154)

50 min · Introductory · No cost



Activity

Think-Pair-Share

List three AI tools you already use. For each, decide whether it predicts, generates, or does both.

Review: Foundations of Generative AI

- Define generative AI
- Explain how generation differs from prediction
- Compare major model families
- Connect applications to business needs



Knowledge check

Question #1: Which statement best distinguishes Generative AI from traditional discriminative ML?

A. It can only process unstructured data

B. It creates new content from learned patterns

C. It does not require training data

D. It always produces deterministic output

Answer #1

Correct answer: B. It creates new content from learned patterns

Generative AI models create new outputs by learning patterns in data.

Question #2: Which model family is widely associated with modern text-generating LLMs?

A. Transformers

B. Decision trees

C. K-means

D. Linear regression

Answer #2

Correct answer: A. Transformers

Modern LLMs primarily use Transformer architectures.

Question #3: What happens during inference?

A. The training dataset is collected

B. Model parameters are initialized

C. A trained model generates an output for an input

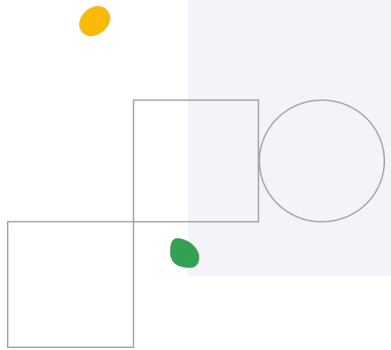
D. Every model weight is retrained

Answer #3

Correct answer: C. A trained model generates an output for an input

Inference is the use of a trained model to produce predictions or generated content.

Questions and discussion



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